

DOMS_with_flocker

Fit DOMs with flocker

Fit dynamic occupancy models with flocker package.

Install / load packages

```
#install.packages("flocker", dependencies = TRUE)
library(flocker)
#install.packages("cmdstanr", repos = c("https://mc-stan.org/r-packages/", getOption("repos")))
library(cmdstanr)
set_cmdstan_path(path = "C:/Users/schifferle1/Documents/cmdstan-2.34.1")
#check_cmdstan_toolchain() # requires RTools installation, see https://mc-stan.org/cmdstanr/
#install_cmdstan(dir = "C:/Users/schifferle1/Documents") # is that right?
library(brms)
library(dplyr)
```

Load data

```
# selected routes and focal years matched to environmental data:
# merged route, year, environment data:
load(file = file.path("data", "route_year_env_data.RData"))

# scale covariates:
route_sel_env_dt_scaled <- route_sel_env_dt_final %>%
  mutate(across(bio2:pr_winter_3yrs, ~ (scale(.)) %>% as.vector()))

# route-year-species information (only surveyed)
load(file = file.path("data", "BBS_for_occ_spec_records.RData")) # output of 1_0_reformat_
```

```
# selected species:
load(file = file.path("data", "final_species_selection.RData")) # output of 1_2_species_se
```

Use House Finch as test species (fitting DOMs with JAGS went well for this).

```
spec <- "House Finch"
```

Assemble data for flocker

```
nyears <- length(unique(route_sel_env_dt_final$Year)) # 25
nsurveys <- 5
nsites <- length(unique(route_sel_env_dt_final$RTENO)) # 476

# species presences:

presences_spec <- bbs_dt_occ %>%
  select(c(English_Common_Name, RTENO, Year, paste0("Count", seq(10, 50, 10)))) %>%
  filter(English_Common_Name == spec)

# match to routes-year-env:

occ_dt_spec <- route_sel_env_dt_scaled %>%
  # add observations:
  collapse::join(presences_spec, on = c("RTENO", "Year"), how = "left") %>%
  # if route was surveyed but species not observed, replace NA with 0:
  mutate(across(Count10:Count50, ~
    case_when(Surveyed == 1 & is.na(.) ~ 0,
              .default = .))) %>%
  # convert bird counts to presence / absence:
  mutate(across(Count10:Count50, ~
    case_when(. > 1 ~ 1,
              .default = .)))

# reformat obs. as array sites x surveys x years (same as for JAGS):

years <- seq(min(occ_dt_spec$Year), max(occ_dt_spec$Year))
y_array <- array(NA, dim = c(nsites, nsurveys, nyears))
for (t in 1:nyears){
  y_array[1:nsites, 1:nsurveys, t] <- as.matrix(occ_dt_spec[which(occ_dt_spec$Year == year
}
```

```

# reformat environmental covariates:

route_sel_env_dt_scaled
env_cov <- vector("list", length = nyears)
for (t in 1:nyears){
  env_cov[[t]] <- route_sel_env_dt_scaled[which(route_sel_env_dt_scaled$Year == years[t]),
}

# covariate for detection probability:

det_cov <- vector("list", length = 1)
names(det_cov) <- "route_section"
det_cov$route_section <- array(NA, dim = c(nsites, nsurveys, nyears))
det_cov$route_section[ , , 1:nyears] <- matrix(rep(c("Sect1", "Sect2", "Sect3", "Sect4", "
# make flocker data:

fd <- make_flocker_data_dynamic(
  obs = y_array,
  unit_covs = env_cov,
  event_covs = det_cov,
  quiet = FALSE
)

```

Fit flocker model

Climate only with default priors:

```

# multi_colex_cl <- flock(
#   f_occ = ~ bio1_3yrs + bio2_3yrs + bio3_3yrs + pr_spring_3yrs + pr_summer_3yrs + pr_aut
#     I(bio1_3yrs^2) + I(bio2_3yrs^2) + I(bio3_3yrs^2) + I(pr_spring_3yrs^2) + I(pr_summer
#   f_det = ~ 1,
#   f_col = ~ bio1 + bio2 + bio3 + pr_spring + pr_summer + pr_autumn + pr_winter + bio15 +
#     I(bio1^2) + I(bio2^2) + I(bio3^2) + I(pr_spring^2) + I(pr_summer^2) + I(pr_autumn^2)
#   f_ex = ~ bio1 + bio2 + bio3 + pr_spring + pr_summer + pr_autumn + pr_winter + bio15 +
#     I(bio1^2) + I(bio2^2) + I(bio3^2) + I(pr_spring^2) + I(pr_summer^2) + I(pr_autumn^2)
#   flocker_data = fd,
#   multiseason = "colex",
#   multi_init = "explicit",
#   backend = "cmdstanr",

```

```
# cores = 3,
# chains = 3
# ) # 54 min
# save(multi_colex_cl, file = file.path("data", paste0(gsub(" ", "_", spec), "_cl4_quadr_S
load(file = file.path("data", paste0(gsub(" ", "_", spec), "_cl4_quadr_Stan.RData")))
multi_colex_cl
```

```
Family: occupancy_multi_colex
Links: mu = identity; occ = identity; colo = identity; ex = identity
Formula: ff_y | vint(ff_n_series, ff_n_unit, ff_n_year, ff_n_rep, ff_Q, ff_unit_index1, ff_u
      occ ~ bio1_3yrs + bio2_3yrs + bio3_3yrs + pr_spring_3yrs + pr_summer_3yrs + pr_autu
      colo ~ bio1 + bio2 + bio3 + pr_spring + pr_summer + pr_autumn + pr_winter + bio15 +
      ex ~ bio1 + bio2 + bio3 + pr_spring + pr_summer + pr_autumn + pr_winter + bio15 + I
Data: data (Number of observations: 56100)
Draws: 3 chains, each with iter = 2000; warmup = 1000; thin = 1;
      total post-warmup draws = 3000
```

Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS
Intercept	-0.08	0.01	-0.11	-0.06	1.00	5632
occ_Intercept	-0.11	0.34	-0.76	0.54	1.00	3403
colo_Intercept	-1.10	0.10	-1.30	-0.91	1.00	3801
ex_Intercept	-1.87	0.11	-2.07	-1.66	1.00	4229
occ_bio1_3yrs	0.65	0.31	0.05	1.28	1.00	2066
occ_bio2_3yrs	-1.11	0.39	-1.88	-0.37	1.00	2145
occ_bio3_3yrs	0.74	0.48	-0.20	1.70	1.00	1919
occ_pr_spring_3yrs	-1.09	0.38	-1.82	-0.33	1.00	2611
occ_pr_summer_3yrs	-0.99	0.30	-1.59	-0.41	1.00	2060
occ_pr_autumn_3yrs	1.35	0.44	0.50	2.23	1.00	1981
occ_pr_winter_3yrs	0.90	0.44	0.01	1.77	1.00	2025
occ_bio15_3yrs	-1.38	0.34	-2.06	-0.73	1.00	2298
occ_Ibio1_3yrsE2	-1.15	0.18	-1.50	-0.81	1.00	3728
occ_Ibio2_3yrsE2	0.23	0.13	-0.03	0.50	1.00	4410
occ_Ibio3_3yrsE2	0.42	0.18	0.06	0.79	1.00	3241
occ_Ipr_spring_3yrsE2	-0.10	0.18	-0.45	0.25	1.00	3242
occ_Ipr_summer_3yrsE2	0.17	0.19	-0.20	0.52	1.00	2360
occ_Ipr_autumn_3yrsE2	0.21	0.16	-0.09	0.53	1.00	2917
occ_Ipr_winter_3yrsE2	-0.49	0.20	-0.91	-0.10	1.00	2082
occ_Ibio15_3yrsE2	0.04	0.14	-0.23	0.30	1.00	2986
colo_bio1	0.09	0.09	-0.08	0.27	1.00	2340
colo_bio2	-0.34	0.11	-0.55	-0.13	1.00	3184
colo_bio3	0.20	0.11	-0.02	0.42	1.00	2319

colo_pr_spring	0.19	0.09	0.03	0.36	1.00	3364
colo_pr_summer	0.09	0.08	-0.05	0.25	1.00	2885
colo_pr_autumn	0.14	0.08	-0.01	0.30	1.00	3780
colo_pr_winter	-0.08	0.09	-0.25	0.09	1.00	3282
colo_bio15	-0.16	0.09	-0.34	0.01	1.00	3517
colo_Ibio1E2	-0.56	0.06	-0.67	-0.45	1.00	4135
colo_Ibio2E2	-0.01	0.05	-0.11	0.09	1.00	4646
colo_Ibio3E2	0.14	0.05	0.04	0.25	1.00	3799
colo_Ipr_springE2	-0.05	0.04	-0.13	0.03	1.00	3477
colo_Ipr_summerE2	-0.02	0.04	-0.11	0.05	1.00	3420
colo_Ipr_autumnE2	-0.15	0.05	-0.26	-0.06	1.00	3679
colo_Ipr_winterE2	-0.01	0.04	-0.09	0.07	1.00	2879
colo_Ibio15E2	0.00	0.03	-0.06	0.07	1.00	2999
ex_bio1	-0.44	0.09	-0.63	-0.26	1.00	2546
ex_bio2	0.62	0.13	0.37	0.88	1.00	2682
ex_bio3	-0.04	0.14	-0.33	0.23	1.00	2219
ex_pr_spring	0.10	0.09	-0.08	0.27	1.00	3232
ex_pr_summer	0.27	0.09	0.10	0.45	1.00	2532
ex_pr_autumn	-0.07	0.09	-0.24	0.11	1.00	2955
ex_pr_winter	-0.47	0.10	-0.67	-0.28	1.00	3338
ex_bio15	0.47	0.11	0.26	0.68	1.00	4399
ex_Ibio1E2	0.36	0.06	0.24	0.48	1.00	4683
ex_Ibio2E2	-0.43	0.07	-0.56	-0.30	1.00	3977
ex_Ibio3E2	-0.35	0.07	-0.50	-0.21	1.00	4468
ex_Ipr_springE2	0.08	0.04	-0.01	0.16	1.00	3218
ex_Ipr_summerE2	-0.04	0.05	-0.15	0.05	1.00	3337
ex_Ipr_autumnE2	0.03	0.04	-0.05	0.11	1.00	3044
ex_Ipr_winterE2	0.32	0.06	0.21	0.43	1.00	2618
ex_Ibio15E2	-0.24	0.06	-0.36	-0.13	1.00	2825
Tail_ESS						
Intercept	2350					
occ_Intercept	2131					
colo_Intercept	2437					
ex_Intercept	2484					
occ_bio1_3yrs	2246					
occ_bio2_3yrs	2215					
occ_bio3_3yrs	2253					
occ_pr_spring_3yrs	2273					
occ_pr_summer_3yrs	2313					
occ_pr_autumn_3yrs	1978					
occ_pr_winter_3yrs	2194					
occ_bio15_3yrs	2473					
occ_Ibio1_3yrsE2	2154					

occ_Ibio2_3yrsE2	2245
occ_Ibio3_3yrsE2	2169
occ_Ipr_spring_3yrsE2	2199
occ_Ipr_summer_3yrsE2	2301
occ_Ipr_autumn_3yrsE2	2504
occ_Ipr_winter_3yrsE2	1997
occ_Ibio15_3yrsE2	2613
colo_bio1	2129
colo_bio2	2358
colo_bio3	2274
colo_pr_spring	2335
colo_pr_summer	2019
colo_pr_autumn	2768
colo_pr_winter	2640
colo_bio15	2282
colo_Ibio1E2	2380
colo_Ibio2E2	2252
colo_Ibio3E2	2457
colo_Ipr_springE2	2489
colo_Ipr_summerE2	2272
colo_Ipr_autumnE2	2357
colo_Ipr_winterE2	2173
colo_Ibio15E2	2398
ex_bio1	2193
ex_bio2	2318
ex_bio3	2126
ex_pr_spring	2747
ex_pr_summer	2304
ex_pr_autumn	2284
ex_pr_winter	2328
ex_bio15	2239
ex_Ibio1E2	2292
ex_Ibio2E2	2319
ex_Ibio3E2	2201
ex_Ipr_springE2	2458
ex_Ipr_summerE2	2340
ex_Ipr_autumnE2	2463
ex_Ipr_winterE2	2284
ex_Ibio15E2	2494

Draws were sampled using `sample(hmc)`. For each parameter, `Bulk_ESS` and `Tail_ESS` are effective sample size measures, and `Rhat` is the potential scale reduction factor on split chains (at convergence, `Rhat` = 1).

Is effective sample size large enough to get stable estimate of uncertainty? -> Bulk_ESS = bulk effective sample size = how well is center of posterior distribution resolved. Tail_ESS = how well are extreme quantiles resolved (?).

```
# extract estimates:
fixef(multi_colex_cl, summary = TRUE, robust = FALSE, probs = c(0.025, 0.975))
```

	Estimate	Est.Error	Q2.5	Q97.5
Intercept	-0.082954446	0.01324485	-0.109334950	-0.05671761
occ_Intercept	-0.108862417	0.33690446	-0.760401850	0.53813045
colo_Intercept	-1.103522054	0.09873855	-1.296945500	-0.91295032
ex_Intercept	-1.866808750	0.10612331	-2.073705500	-1.66314575
occ_bio1_3yrs	0.649507077	0.31213508	0.049073320	1.28097475
occ_bio2_3yrs	-1.106967062	0.38649278	-1.884742750	-0.37146727
occ_bio3_3yrs	0.737256872	0.47994728	-0.196587750	1.70435175
occ_pr_spring_3yrs	-1.085096199	0.38255646	-1.821522750	-0.33272932
occ_pr_summer_3yrs	-0.988680156	0.30332678	-1.587203000	-0.40534677
occ_pr_autumn_3yrs	1.348512235	0.44426477	0.499432850	2.23029625
occ_pr_winter_3yrs	0.895113420	0.44136350	0.012450340	1.77218350
occ_bio15_3yrs	-1.375876985	0.34260251	-2.056057000	-0.72687117
occ_Ibio1_3yrsE2	-1.145414354	0.17603739	-1.502061250	-0.81284353
occ_Ibio2_3yrsE2	0.229532331	0.13251050	-0.030457862	0.49501535
occ_Ibio3_3yrsE2	0.415194380	0.18338072	0.060428348	0.78921060
occ_Ipr_spring_3yrsE2	-0.103574475	0.17879713	-0.452462475	0.24894678
occ_Ipr_summer_3yrsE2	0.167649170	0.18544558	-0.199851725	0.52171943
occ_Ipr_autumn_3yrsE2	0.208183253	0.16234916	-0.093051515	0.53069102
occ_Ipr_winter_3yrsE2	-0.487989822	0.20263088	-0.914619025	-0.09998672
occ_Ibio15_3yrsE2	0.041719525	0.13602714	-0.229293725	0.30417983
colo_bio1	0.092185868	0.08944733	-0.081848647	0.26684915
colo_bio2	-0.342316932	0.10754944	-0.554214575	-0.12919977
colo_bio3	0.198152259	0.11221593	-0.023961365	0.42111635
colo_pr_spring	0.194993898	0.08545290	0.029200770	0.36334870
colo_pr_summer	0.093226189	0.07513213	-0.050918450	0.24754488
colo_pr_autumn	0.139394614	0.07909314	-0.008074587	0.29644365
colo_pr_winter	-0.080990247	0.08645152	-0.250366425	0.08771059
colo_bio15	-0.159733858	0.08908208	-0.337835975	0.01481243
colo_Ibio1E2	-0.555254609	0.05620080	-0.665491825	-0.44583030
colo_Ibio2E2	-0.008261647	0.05163718	-0.110621875	0.09111744
colo_Ibio3E2	0.143279371	0.05362134	0.039712952	0.24926410
colo_Ipr_springE2	-0.051505918	0.04039705	-0.134344100	0.02507766
colo_Ipr_summerE2	-0.024408660	0.04108502	-0.106248800	0.05409516
colo_Ipr_autumnE2	-0.150451797	0.05120471	-0.255435550	-0.05529011

colo_Ipr_winterE2	-0.006154458	0.04054023	-0.091308817	0.06983835
colo_Ibio15E2	0.004906791	0.03340473	-0.061905417	0.06861623
ex_bio1	-0.444221597	0.09412628	-0.626712000	-0.26119657
ex_bio2	0.624449914	0.13331447	0.366006550	0.88044643
ex_bio3	-0.041521738	0.14463897	-0.329911125	0.23350893
ex_pr_spring	0.096885298	0.09132961	-0.081378820	0.27302673
ex_pr_summer	0.270955561	0.08965366	0.095429357	0.45072075
ex_pr_autumn	-0.065109814	0.09029176	-0.242287325	0.11376400
ex_pr_winter	-0.471926760	0.09869004	-0.670851325	-0.28151165
ex_bio15	0.468457830	0.10565210	0.261351500	0.67646100
ex_Ibio1E2	0.360407304	0.06053883	0.239495775	0.48059010
ex_Ibio2E2	-0.429753140	0.06549517	-0.559108525	-0.30392002
ex_Ibio3E2	-0.349976951	0.07331195	-0.500027300	-0.21106282
ex_Ipr_springE2	0.078627925	0.04350020	-0.010981850	0.15960550
ex_Ipr_summerE2	-0.043067224	0.05122914	-0.149698675	0.05441057
ex_Ipr_autumnE2	0.033837079	0.04244058	-0.054702305	0.11325833
ex_Ipr_winterE2	0.320577719	0.05592597	0.210975225	0.43077923
ex_Ibio15E2	-0.241123488	0.05759028	-0.356519250	-0.13146422

Get estimates for occupancy, extinction and colonisation probability:

For each site and year and survey (route section) we get mean and quantiles:

```
# posterior predictive distribution for the modeled probabilities (occ., det., col., ext.)
# takes a while (20 min?)
# posterior_preds <- fitted_flocker(
#   multi_colex_cl,
#   components = c("occ", "det", "col", "ex"),
#   new_data = NULL,
#   unit_level = FALSE,
#   summarise = TRUE,
#   CI = c(0.025, 0.975),
#   draw_ids = NULL,
#   response = TRUE, # response or logit scale
#   re_formula = NULL,
#   allow_new_levels = FALSE,
#   sample_new_levels = "uncertainty"
# )
#save(posterior_preds, file = file.path("data", paste0(gsub(" ", "_", spec), "_cl4_quadra_SDa
load(file = file.path("data", paste0(gsub(" ", "_", spec), "_cl4_quadra_Stan_post_preds.RDa
str(posterior_preds)
```


List of 4

```
$ linpred_occ: num [1:476, 1:5, 1:25, 1:3] 0.00822 0.10817 0.1149 0.01719 0.05429 ...
..- attr(*, "dimnames")=List of 4
.. ..$ : chr [1:476] "site_1" "site_2" "site_3" "site_4" ...
.. ..$ : chr [1:5] "visit_1" "visit_2" "visit_3" "visit_4" ...
.. ..$ : chr [1:25] "timestep_1" "timestep_2" "timestep_3" "timestep_4" ...
.. ..$ : chr [1:3] "mean" "Q2.5" "Q97.5"
$ linpred_col: num [1:476, 1:5, 1:25, 1:3] 0.0929 0.165 0.1515 0.0759 0.1474 ...
..- attr(*, "dimnames")=List of 4
.. ..$ : chr [1:476] "site_1" "site_2" "site_3" "site_4" ...
.. ..$ : chr [1:5] "visit_1" "visit_2" "visit_3" "visit_4" ...
.. ..$ : chr [1:25] "timestep_1" "timestep_2" "timestep_3" "timestep_4" ...
.. ..$ : chr [1:3] "mean" "Q2.5" "Q97.5"
$ linpred_ex : num [1:476, 1:5, 1:25, 1:3] 0.59 0.37 0.231 0.567 0.2 ...
..- attr(*, "dimnames")=List of 4
.. ..$ : chr [1:476] "site_1" "site_2" "site_3" "site_4" ...
.. ..$ : chr [1:5] "visit_1" "visit_2" "visit_3" "visit_4" ...
.. ..$ : chr [1:25] "timestep_1" "timestep_2" "timestep_3" "timestep_4" ...
.. ..$ : chr [1:3] "mean" "Q2.5" "Q97.5"
$ linpred_det: num [1:476, 1:5, 1:25, 1:3] 0.479 0.479 0.479 0.479 0.479 ...
..- attr(*, "dimnames")=List of 4
.. ..$ : chr [1:476] "site_1" "site_2" "site_3" "site_4" ...
.. ..$ : chr [1:5] "visit_1" "visit_2" "visit_3" "visit_4" ...
.. ..$ : chr [1:25] "timestep_1" "timestep_2" "timestep_3" "timestep_4" ...
.. ..$ : chr [1:3] "mean" "Q2.5" "Q97.5"
```

```
posterior_preds$linpred_occ[1:10, , 1, ]
```

, , mean

	visit_1	visit_2	visit_3	visit_4	visit_5
site_1	0.008221026	0.008221026	0.008221026	0.008221026	0.008221026
site_2	0.108171000	0.108171000	0.108171000	0.108171000	0.108171000
site_3	0.114902563	0.114902563	0.114902563	0.114902563	0.114902563
site_4	0.017188359	0.017188359	0.017188359	0.017188359	0.017188359
site_5	0.054291965	0.054291965	0.054291965	0.054291965	0.054291965
site_6	0.203589501	0.203589501	0.203589501	0.203589501	0.203589501
site_7	0.041899753	0.041899753	0.041899753	0.041899753	0.041899753
site_8	0.011406283	0.011406283	0.011406283	0.011406283	0.011406283
site_9	0.025184780	0.025184780	0.025184780	0.025184780	0.025184780
site_10	0.017740666	0.017740666	0.017740666	0.017740666	0.017740666

, , Q2.5

	visit_1	visit_2	visit_3	visit_4	visit_5
site_1	0.0007281831	0.0007281831	0.0007281831	0.0007281831	0.0007281831
site_2	0.0259409993	0.0259409993	0.0259409993	0.0259409993	0.0259409993
site_3	0.0416066614	0.0416066614	0.0416066614	0.0416066614	0.0416066614
site_4	0.0017544480	0.0017544480	0.0017544480	0.0017544480	0.0017544480
site_5	0.0154470171	0.0154470171	0.0154470171	0.0154470171	0.0154470171
site_6	0.0934887064	0.0934887064	0.0934887064	0.0934887064	0.0934887064
site_7	0.0122337527	0.0122337527	0.0122337527	0.0122337527	0.0122337527
site_8	0.0013886466	0.0013886466	0.0013886466	0.0013886466	0.0013886466
site_9	0.0049984203	0.0049984203	0.0049984203	0.0049984203	0.0049984203
site_10	0.0022080650	0.0022080650	0.0022080650	0.0022080650	0.0022080650

, , Q97.5

	visit_1	visit_2	visit_3	visit_4	visit_5
site_1	0.02974116	0.02974116	0.02974116	0.02974116	0.02974116
site_2	0.26074313	0.26074313	0.26074313	0.26074313	0.26074313
site_3	0.22331818	0.22331818	0.22331818	0.22331818	0.22331818
site_4	0.06144262	0.06144262	0.06144262	0.06144262	0.06144262
site_5	0.12253877	0.12253877	0.12253877	0.12253877	0.12253877
site_6	0.35977817	0.35977817	0.35977817	0.35977817	0.35977817
site_7	0.09612020	0.09612020	0.09612020	0.09612020	0.09612020
site_8	0.03893006	0.03893006	0.03893006	0.03893006	0.03893006
site_9	0.07031825	0.07031825	0.07031825	0.07031825	0.07031825
site_10	0.05583060	0.05583060	0.05583060	0.05583060	0.05583060

Modify priors:

Flocker gives us default priors for a model formula:

```
get_flocker_prior(f_occ = ~ bio1_3yrs + bio2_3yrs + bio3_3yrs + pr_spring_3yrs + pr_summer_3yrs +
  I(bio1_3yrs^2) + I(bio2_3yrs^2) + I(bio3_3yrs^2) + I(pr_spring_3yrs^2) + I(pr_summer_3yrs^2),
f_det = ~ 1,
f_col = ~ bio1 + bio2 + bio3 + pr_spring + pr_summer + pr_autumn + pr_winter + bio15 +
  I(bio1^2) + I(bio2^2) + I(bio3^2) + I(pr_spring^2) + I(pr_summer^2) + I(pr_autumn^2) + I(pr_winter^2),
f_ex = ~ bio1 + bio2 + bio3 + pr_spring + pr_summer + pr_autumn + pr_winter + bio15 +
  I(bio1^2) + I(bio2^2) + I(bio3^2) + I(pr_spring^2) + I(pr_summer^2) + I(pr_autumn^2) + I(pr_winter^2),
flocker_data = fd,
multiseason = "colex",
```

```
multi_init = "explicit")
```

	prior	class	coef	group	resp	dpar	nlpar	lb	ub
student_t(3, 0, 2.5)		Intercept							
	(flat)	b				colo			
	(flat)	b	bio1			colo			
	(flat)	b	bio15			colo			
	(flat)	b	bio2			colo			
	(flat)	b	bio3			colo			
	(flat)	b	Ibio15E2			colo			
	(flat)	b	Ibio1E2			colo			
	(flat)	b	Ibio2E2			colo			
	(flat)	b	Ibio3E2			colo			
	(flat)	b	Ipr_autumnE2			colo			
	(flat)	b	Ipr_springE2			colo			
	(flat)	b	Ipr_summerE2			colo			
	(flat)	b	Ipr_winterE2			colo			
	(flat)	b	pr_autumn			colo			
	(flat)	b	pr_spring			colo			
	(flat)	b	pr_summer			colo			
	(flat)	b	pr_winter			colo			
	(flat)	Intercept				colo			
	(flat)	b				ex			
	(flat)	b	bio1			ex			
	(flat)	b	bio15			ex			
	(flat)	b	bio2			ex			
	(flat)	b	bio3			ex			
	(flat)	b	Ibio15E2			ex			
	(flat)	b	Ibio1E2			ex			
	(flat)	b	Ibio2E2			ex			
	(flat)	b	Ibio3E2			ex			
	(flat)	b	Ipr_autumnE2			ex			
	(flat)	b	Ipr_springE2			ex			
	(flat)	b	Ipr_summerE2			ex			
	(flat)	b	Ipr_winterE2			ex			
	(flat)	b	pr_autumn			ex			
	(flat)	b	pr_spring			ex			
	(flat)	b	pr_summer			ex			
	(flat)	b	pr_winter			ex			
	(flat)	Intercept				ex			
	(flat)	b				occ			
	(flat)	b	bio1_3yrs			occ			

(flat)	b	bio15_3yrs	occ
(flat)	b	bio2_3yrs	occ
(flat)	b	bio3_3yrs	occ
(flat)	b	Ibio1_3yrsE2	occ
(flat)	b	Ibio15_3yrsE2	occ
(flat)	b	Ibio2_3yrsE2	occ
(flat)	b	Ibio3_3yrsE2	occ
(flat)	b	Ipr_autumn_3yrsE2	occ
(flat)	b	Ipr_spring_3yrsE2	occ
(flat)	b	Ipr_summer_3yrsE2	occ
(flat)	b	Ipr_winter_3yrsE2	occ
(flat)	b	pr_autumn_3yrs	occ
(flat)	b	pr_spring_3yrs	occ
(flat)	b	pr_summer_3yrs	occ
(flat)	b	pr_winter_3yrs	occ
(flat)		Intercept	occ
source			
default			
default			
(vectorized)			
(vectorized)			
(vectorized)			
(vectorized)			
(vectorized)			
(vectorized)			
(vectorized)			
(vectorized)			
(vectorized)			
(vectorized)			
(vectorized)			
(vectorized)			
(vectorized)			
(vectorized)			
(vectorized)			
(vectorized)			
default			
default			
(vectorized)			
(vectorized)			
(vectorized)			
(vectorized)			
(vectorized)			
(vectorized)			

```

(vectorized)
(vectorized)
(vectorized)
(vectorized)
(vectorized)
(vectorized)
(vectorized)
(vectorized)
(vectorized)
      default
      default
(vectorized)
(vectorized)
(vectorized)
(vectorized)
(vectorized)
(vectorized)
(vectorized)
(vectorized)
(vectorized)
(vectorized)
(vectorized)
(vectorized)
(vectorized)
(vectorized)
(vectorized)
      default

```

To change priors we use the brms implementation. Brms includes two shrinkage/ regularization priors, horseshoe and R2D2.

- horseshoe:
 - “Horseshoe prior is symmetric around zero with fat tails and an infinitely large spike at zero. This makes it ideal for sparse models that have many regression coefficients, although only a minority of them is non-zero.”
 - uses two levels of regularization, global and local, the global parameter τ_0 pulls all the weights globally towards zero, while the thick half-Cauchy tails for the local scales τ_j allow some of the weights to escape the shrinkage
 - brms arguments (regularized horseshoe):

- * df (default 1): Degrees of freedom of student-t prior of the local shrinkage parameters
- * scale_global (default 1): Scale of the student-t prior of the global shrinkage parameter; determines how much shrinkage is applied, Piironen and Vehtari (2017) recommend to specifying the ratio of the expected number of non-zero coefficients to the expected number of zero coefficients 'par_ratio' rather than 'scale_global' directly
- * df_global (default 1): Degrees of freedom of student-t prior of the global shrinkage parameter
- * scale_slab (default = 2): scale of student-t-slab (regularization, only affects non-zero coeff., expected value of the non-zero coefficient?)
- * df_slab (default = 4): df of the student-t-slab (regularization, only affects non-zero coeff.)
- * par_ratio: Ratio of the expected number of non-zero coefficients to the expected number of zero coefficients; specified, `scale_global` is ignored and internally computed as `par_ratio / sqrt(N)`, N is the total number of observations in the data

For now, I defined `par_ratio` instead of `scale_global`. I assumed that roughly 1/4 of the coefficients may be non-zero -> `par_ratio = (1/4)/(3/4) = 0.33`. This results in a `scale_global` of `0.3/sqrt(11900) = 0.003` (default is 1). Not sure about this.

“Specifying priors for parameters in formulas other than detection can be done with reference to the `dpar` column.”

```
# multi_colex_all_cl_prior <- flock(
#   f_occ = ~ bio1_3yrs + bio2_3yrs + bio3_3yrs + pr_spring_3yrs + pr_summer_3yrs + pr_aut
#     I(bio1_3yrs^2) + I(bio2_3yrs^2) + I(bio3_3yrs^2) + I(pr_spring_3yrs^2) + I(pr_summer
#   f_det = ~ 1,
#   f_col = ~ bio1 + bio2 + bio3 + pr_spring + pr_summer + pr_autumn + pr_winter + bio15 +
#     I(bio1^2) + I(bio2^2) + I(bio3^2) + I(pr_spring^2) + I(pr_summer^2) + I(pr_autumn^2)
#   f_ex = ~ bio1 + bio2 + bio3 + pr_spring + pr_summer + pr_autumn + pr_winter + bio15 +
#     I(bio1^2) + I(bio2^2) + I(bio3^2) + I(pr_spring^2) + I(pr_summer^2) + I(pr_autumn^2)
#   flocker_data = fd2_det,
#   prior = c(brms::set_prior("logistic(0,1)", class = "Intercept") + # flat on probabilit
#     brms::set_prior("logistic(0,1)", class = "Intercept", dpar = "occ"),
#     brms::set_prior("logistic(0,1)", class = "Intercept", dpar = "colo"),
#     brms::set_prior("logistic(0,1)", class = "Intercept", dpar = "ex"),
#     brms::set_prior("horseshoe(df = 1, par_ratio = 0.3)", dpar = "occ", class =
#     brms::set_prior("horseshoe(df = 1, par_ratio = 0.3)", dpar = "colo", class =
```

```
#           brms::set_prior("horseshoe(df = 1, par_ratio = 0.3)", dpar = "ex", class = "
# multiseason = "colex",
# multi_init = "explicit",
# backend = "cmdstanr",
# cores = 3,
# chains = 3
# )# 11 of 3000 transitions divergent, 12000s = 3
# save(multi_colex_all_cl_prior, file = file.path("data", paste0(gsub(" ", "_", spec), "_c
load(file = file.path("data", paste0(gsub(" ", "_", spec), "_cl4_quad_rStan_prior.RData"))
multi_colex_all_cl_prior
```

Warning: There were 11 divergent transitions after warmup. Increasing
adapt_delta above 0.8 may help. See
<http://mc-stan.org/misc/warnings.html#divergent-transitions-after-warmup>

Family: occupancy_multi_colex
Links: mu = identity; occ = identity; colo = identity; ex = identity
Formula: ff_y | vint(ff_n_series, ff_n_unit, ff_n_year, ff_n_rep, ff_Q, ff_unit_index1, ff_u
occ ~ bio1_3yrs + bio2_3yrs + bio3_3yrs + pr_spring_3yrs + pr_summer_3yrs + pr_autu
colo ~ bio1 + bio2 + bio3 + pr_spring + pr_summer + pr_autumn + pr_winter + bio15 +
ex ~ bio1 + bio2 + bio3 + pr_spring + pr_summer + pr_autumn + pr_winter + bio15 + I
Data: data (Number of observations: 56100)
Draws: 3 chains, each with iter = 2000; warmup = 1000; thin = 1;
total post-warmup draws = 3000

Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS
Intercept	-0.08	0.01	-0.11	-0.06	1.00	3937
occ_Intercept	-0.07	0.32	-0.68	0.57	1.00	2572
colo_Intercept	-1.15	0.09	-1.33	-0.97	1.00	2893
ex_Intercept	-1.86	0.10	-2.06	-1.66	1.00	2895
occ_bio1_3yrs	0.40	0.29	-0.07	0.96	1.00	1796
occ_bio2_3yrs	-1.01	0.36	-1.72	-0.35	1.00	1934
occ_bio3_3yrs	0.71	0.44	-0.07	1.55	1.00	1730
occ_pr_spring_3yrs	-0.89	0.38	-1.65	-0.13	1.00	2153
occ_pr_summer_3yrs	-0.73	0.31	-1.33	-0.10	1.00	2092
occ_pr_autumn_3yrs	1.04	0.43	0.18	1.89	1.00	2412
occ_pr_winter_3yrs	0.78	0.45	-0.03	1.66	1.00	1578
occ_bio15_3yrs	-1.13	0.32	-1.75	-0.50	1.00	2612
occ_Ibio1_3yrsE2	-1.03	0.17	-1.36	-0.71	1.00	2698
occ_Ibio2_3yrsE2	0.21	0.13	-0.01	0.46	1.00	1931

occ_Ibio3_3yrsE2	0.34	0.19	-0.01	0.71	1.00	1686
occ_Ipr_spring_3yrsE2	-0.04	0.14	-0.35	0.21	1.00	2574
occ_Ipr_summer_3yrsE2	0.11	0.15	-0.16	0.44	1.00	2343
occ_Ipr_autumn_3yrsE2	0.18	0.15	-0.06	0.50	1.00	2475
occ_Ipr_winter_3yrsE2	-0.43	0.20	-0.81	-0.04	1.00	1928
occ_Ibio15_3yrsE2	0.04	0.10	-0.17	0.26	1.00	2473
colo_biol	0.11	0.08	-0.02	0.26	1.00	1304
colo_biol2	-0.22	0.11	-0.44	-0.00	1.00	1364
colo_biol3	0.04	0.07	-0.05	0.23	1.00	1305
colo_pr_spring	0.09	0.08	-0.01	0.26	1.00	1644
colo_pr_summer	0.06	0.06	-0.03	0.21	1.00	1960
colo_pr_autumn	0.11	0.09	-0.01	0.29	1.00	1469
colo_pr_winter	-0.00	0.04	-0.09	0.09	1.00	2293
colo_biol15	-0.04	0.06	-0.19	0.05	1.00	1403
colo_Ibio1E2	-0.51	0.06	-0.63	-0.40	1.00	2204
colo_Ibio2E2	0.00	0.03	-0.06	0.07	1.00	2834
colo_Ibio3E2	0.08	0.06	-0.01	0.20	1.00	1214
colo_Ipr_springE2	-0.02	0.03	-0.09	0.03	1.00	2236
colo_Ipr_summerE2	-0.00	0.02	-0.06	0.04	1.00	2637
colo_Ipr_autumnE2	-0.12	0.05	-0.23	-0.01	1.00	2124
colo_Ipr_winterE2	-0.01	0.03	-0.07	0.03	1.00	2588
colo_Ibio15E2	-0.01	0.02	-0.06	0.02	1.00	2486
ex_biol	-0.41	0.08	-0.57	-0.24	1.00	2754
ex_biol2	0.59	0.12	0.37	0.83	1.00	2683
ex_biol3	-0.05	0.10	-0.30	0.13	1.00	3394
ex_pr_spring	0.09	0.09	-0.04	0.27	1.00	2305
ex_pr_summer	0.21	0.09	0.04	0.39	1.00	2078
ex_pr_autumn	-0.02	0.06	-0.17	0.10	1.00	2679
ex_pr_winter	-0.45	0.09	-0.63	-0.26	1.00	2870
ex_biol15	0.43	0.10	0.22	0.62	1.00	2503
ex_Ibio1E2	0.34	0.06	0.22	0.46	1.00	2986
ex_Ibio2E2	-0.41	0.06	-0.53	-0.28	1.00	2871
ex_Ibio3E2	-0.33	0.07	-0.48	-0.20	1.00	3173
ex_Ipr_springE2	0.07	0.04	-0.01	0.15	1.00	1968
ex_Ipr_summerE2	-0.02	0.04	-0.11	0.05	1.00	2468
ex_Ipr_autumnE2	0.02	0.03	-0.04	0.09	1.00	2372
ex_Ipr_winterE2	0.29	0.06	0.19	0.40	1.00	2846
ex_Ibio15E2	-0.21	0.06	-0.32	-0.10	1.00	2952
Tail_ESS						
Intercept	2152					
occ_Intercept	2583					
colo_Intercept	2907					
ex_Intercept	2765					

occ_bio1_3yrs	2152
occ_bio2_3yrs	2016
occ_bio3_3yrs	1144
occ_pr_spring_3yrs	1573
occ_pr_summer_3yrs	1771
occ_pr_autumn_3yrs	1639
occ_pr_winter_3yrs	1325
occ_bio15_3yrs	2591
occ_Ibio1_3yrsE2	2631
occ_Ibio2_3yrsE2	1533
occ_Ibio3_3yrsE2	1158
occ_Ipr_spring_3yrsE2	2739
occ_Ipr_summer_3yrsE2	2511
occ_Ipr_autumn_3yrsE2	2454
occ_Ipr_winter_3yrsE2	1758
occ_Ibio15_3yrsE2	2920
colo_bio1	1332
colo_bio2	868
colo_bio3	2708
colo_pr_spring	2590
colo_pr_summer	2554
colo_pr_autumn	1801
colo_pr_winter	1233
colo_bio15	2342
colo_Ibio1E2	2327
colo_Ibio2E2	2653
colo_Ibio3E2	1303
colo_Ipr_springE2	2741
colo_Ipr_summerE2	2682
colo_Ipr_autumnE2	1953
colo_Ipr_winterE2	2770
colo_Ibio15E2	1869
ex_bio1	2628
ex_bio2	2561
ex_bio3	2837
ex_pr_spring	2274
ex_pr_summer	1624
ex_pr_autumn	2852
ex_pr_winter	2884
ex_bio15	2364
ex_Ibio1E2	2832
ex_Ibio2E2	2547
ex_Ibio3E2	2576

ex_Ipr_springE2	1574
ex_Ipr_summerE2	2679
ex_Ipr_autumnE2	2422
ex_Ipr_winterE2	2592
ex_Ibio15E2	2362

Draws were sampled using `sample(hmc)`. For each parameter, `Bulk_ESS` and `Tail_ESS` are effective sample size measures, and `Rhat` is the potential scale reduction factor on split chains (at convergence, `Rhat` = 1).

Compare to same model in JAGS using lasso regularization:

```
library(jagsUI)
load(file.path("results", "House_Finch_cl_alt4_quadr_lasso_GoF_inits.RData"))

jags_estimates <- out$summary[which(grepl("(alpha)|(beta)", rownames(out$summary))), c("me

stan_estimates <- fixef(multi_colex_all_cl_prior, summary = TRUE, robust = FALSE, probs =

jags_reformatted <- rbind(c(NA, NA, NA),
  jags_estimates["alpha_psi", c("mean", "2.5%", "97.5%")],
  jags_estimates["alpha_gamma", c("mean", "2.5%", "97.5%")],
  jags_estimates["alpha_eps", c("mean", "2.5%", "97.5%")],
  jags_estimates["beta_psi[1]", c("mean", "2.5%", "97.5%")],
  jags_estimates["beta_psi[2]", c("mean", "2.5%", "97.5%")],
  jags_estimates["beta_psi[3]", c("mean", "2.5%", "97.5%")],
  jags_estimates["beta_psi[4]", c("mean", "2.5%", "97.5%")],
  jags_estimates["beta_psi[5]", c("mean", "2.5%", "97.5%")],
  jags_estimates["beta_psi[6]", c("mean", "2.5%", "97.5%")],
  jags_estimates["beta_psi[7]", c("mean", "2.5%", "97.5%")],
  jags_estimates["beta_psi[8]", c("mean", "2.5%", "97.5%")],
  jags_estimates["beta_psi[9]", c("mean", "2.5%", "97.5%")],
  jags_estimates["beta_psi[10]", c("mean", "2.5%", "97.5%")],
  jags_estimates["beta_psi[11]", c("mean", "2.5%", "97.5%")],
  jags_estimates["beta_psi[12]", c("mean", "2.5%", "97.5%")],
  jags_estimates["beta_psi[13]", c("mean", "2.5%", "97.5%")],
  jags_estimates["beta_psi[14]", c("mean", "2.5%", "97.5%")],
  jags_estimates["beta_psi[15]", c("mean", "2.5%", "97.5%")],
  jags_estimates["beta_psi[16]", c("mean", "2.5%", "97.5%")],
  jags_estimates["beta_gamma[1]", c("mean", "2.5%", "97.5%")],
  jags_estimates["beta_gamma[2]", c("mean", "2.5%", "97.5%")],
```

```

jags_estimates["beta_gamma[3]", c("mean", "2.5%", "97.5%")],
jags_estimates["beta_gamma[4]", c("mean", "2.5%", "97.5%")],
jags_estimates["beta_gamma[5]", c("mean", "2.5%", "97.5%")],
jags_estimates["beta_gamma[6]", c("mean", "2.5%", "97.5%")],
jags_estimates["beta_gamma[7]", c("mean", "2.5%", "97.5%")],
jags_estimates["beta_gamma[8]", c("mean", "2.5%", "97.5%")],
jags_estimates["beta_gamma[9]", c("mean", "2.5%", "97.5%")],
jags_estimates["beta_gamma[10]", c("mean", "2.5%", "97.5%")],
jags_estimates["beta_gamma[11]", c("mean", "2.5%", "97.5%")],
jags_estimates["beta_gamma[12]", c("mean", "2.5%", "97.5%")],
jags_estimates["beta_gamma[13]", c("mean", "2.5%", "97.5%")],
jags_estimates["beta_gamma[14]", c("mean", "2.5%", "97.5%")],
jags_estimates["beta_gamma[15]", c("mean", "2.5%", "97.5%")],
jags_estimates["beta_gamma[16]", c("mean", "2.5%", "97.5%")],
jags_estimates["beta_eps[1]", c("mean", "2.5%", "97.5%")],
jags_estimates["beta_eps[2]", c("mean", "2.5%", "97.5%")],
jags_estimates["beta_eps[3]", c("mean", "2.5%", "97.5%")],
jags_estimates["beta_eps[4]", c("mean", "2.5%", "97.5%")],
jags_estimates["beta_eps[5]", c("mean", "2.5%", "97.5%")],
jags_estimates["beta_eps[6]", c("mean", "2.5%", "97.5%")],
jags_estimates["beta_eps[7]", c("mean", "2.5%", "97.5%")],
jags_estimates["beta_eps[8]", c("mean", "2.5%", "97.5%")],
jags_estimates["beta_eps[9]", c("mean", "2.5%", "97.5%")],
jags_estimates["beta_eps[10]", c("mean", "2.5%", "97.5%")],
jags_estimates["beta_eps[11]", c("mean", "2.5%", "97.5%")],
jags_estimates["beta_eps[12]", c("mean", "2.5%", "97.5%")],
jags_estimates["beta_eps[13]", c("mean", "2.5%", "97.5%")],
jags_estimates["beta_eps[14]", c("mean", "2.5%", "97.5%")],
jags_estimates["beta_eps[15]", c("mean", "2.5%", "97.5%")],
jags_estimates["beta_eps[16]", c("mean", "2.5%", "97.5%")])

```

```

comparison <- cbind(as.data.frame(stan_estimates)[,c(1,3,4)], jags_reformatted)
colnames(comparison) <- c("Stan_mean", "Stan_2.5%", "Stan_97.5%",
                          "JAGS_mean", "JAGS_2.5%", "JAGS_97.5%")
comparison <- comparison[,c(1,4,2,5,3,6)]
comparison

```

	Stan_mean	JAGS_mean	Stan_2.5%	JAGS_2.5%
Intercept	-0.0822452383	NA	-0.107857825	NA
occ_Intercept	-0.0683503039	-0.0642274810	-0.679089075	-0.685412000
colo_Intercept	-1.1527798403	-1.1362771998	-1.331168750	-1.322507081
ex_Intercept	-1.8595954733	-1.8659996264	-2.055140500	-2.057516437

occ_bio1_3yrs	0.4020563680	0.4368768704	-0.068786320	-0.089095159
occ_bio2_3yrs	-1.0141307709	-0.9500574567	-1.720864000	-1.593167968
occ_bio3_3yrs	0.7087486704	0.6261029185	-0.066211317	-0.075993420
occ_pr_spring_3yrs	-0.8881981362	-0.8691551312	-1.650127250	-1.542486118
occ_pr_summer_3yrs	-0.7313735452	-0.7067892324	-1.325740750	-1.233693936
occ_pr_autumn_3yrs	1.0390351432	0.9371098809	0.180181075	0.237099789
occ_pr_winter_3yrs	0.7758141760	0.8647625425	-0.030811662	0.157990969
occ_bio15_3yrs	-1.1347516271	-1.0951443003	-1.750982750	-1.715651919
occ_Ibio1_3yrsE2	-1.0280097417	-1.0057216035	-1.358025250	-1.334674315
occ_Ibio2_3yrsE2	0.2078614577	0.2289858596	-0.013761145	0.004016017
occ_Ibio3_3yrsE2	0.3356029871	0.3329181705	-0.005850204	0.019718687
occ_Ipr_spring_3yrsE2	-0.0426722352	-0.0733635168	-0.351493625	-0.371053872
occ_Ipr_summer_3yrsE2	0.1100460845	0.0944082202	-0.160271550	-0.211073834
occ_Ipr_autumn_3yrsE2	0.1820573704	0.2365345495	-0.056559382	-0.047738873
occ_Ipr_winter_3yrsE2	-0.4317592122	-0.4879985904	-0.812022725	-0.894670012
occ_Ibio15_3yrsE2	0.0372678306	0.0389320907	-0.170418100	-0.179327899
colo_bio1	0.1081510049	0.1071595387	-0.015549062	-0.024214572
colo_bio2	-0.2234209563	-0.2217036057	-0.435318525	-0.410108577
colo_bio3	0.0444780556	0.0771049654	-0.054253677	-0.068035065
colo_pr_spring	0.0894741175	0.1364932686	-0.013896973	0.002120507
colo_pr_summer	0.0629649038	0.0835164447	-0.025695953	-0.033371062
colo_pr_autumn	0.1090749431	0.1172116443	-0.013112805	-0.010344838
colo_pr_winter	-0.0007842717	0.0007255593	-0.092883922	-0.127268891
colo_bio15	-0.0384324655	-0.0820194530	-0.193310775	-0.251247728
colo_Ibio1E2	-0.5128343650	-0.5078556481	-0.625891425	-0.621052636
colo_Ibio2E2	0.0022683295	0.0010774427	-0.060335337	-0.082260834
colo_Ibio3E2	0.0840976052	0.1051077951	-0.006472069	0.009565338
colo_Ipr_springE2	-0.0178799970	-0.0360971465	-0.090784035	-0.113585171
colo_Ipr_summerE2	-0.0042453281	-0.0173106406	-0.060700588	-0.091326959
colo_Ipr_autumnE2	-0.1182382431	-0.1274082877	-0.228995775	-0.223291012
colo_Ipr_winterE2	-0.0143257085	-0.0167448657	-0.072558120	-0.082436706
colo_Ibio15E2	-0.0123557653	-0.0078924610	-0.061992483	-0.067121736
ex_bio1	-0.4116297687	-0.4046832303	-0.569738525	-0.561535221
ex_bio2	0.5895844320	0.5834128218	0.365375950	0.352899114
ex_bio3	-0.0476740761	-0.0503657790	-0.295437300	-0.278990356
ex_pr_spring	0.0913183776	0.1019593922	-0.043756605	-0.054581609
ex_pr_summer	0.2125295578	0.2362300883	0.043123895	0.060782069
ex_pr_autumn	-0.0197370521	-0.0529780774	-0.165254650	-0.225067333
ex_pr_winter	-0.4463375037	-0.4289624197	-0.625917850	-0.616606499
ex_bio15	0.4278092518	0.4147701142	0.223658625	0.211672189
ex_Ibio1E2	0.3449525977	0.3522712342	0.217737600	0.234766197
ex_Ibio2E2	-0.4065271580	-0.4033344498	-0.532548175	-0.526780824
ex_Ibio3E2	-0.3348525256	-0.3339459640	-0.483584250	-0.471958615

ex_Ipr_springE2	0.0660707080	0.0663329042	-0.010882295	-0.012877787
ex_Ipr_summerE2	-0.0190483372	-0.0337968013	-0.109539125	-0.135759649
ex_Ipr_autumnE2	0.0189109918	0.0309493927	-0.040804910	-0.046194285
ex_Ipr_winterE2	0.2906655569	0.2839284923	0.185373000	0.178536633
ex_Ibio15E2	-0.2095685288	-0.2051275843	-0.322739875	-0.327736647
	Stan_97.5%	JAGS_97.5%		
Intercept	-0.057374790		NA	
occ_Intercept	0.571708675	0.56856622		
colo_Intercept	-0.965956275	-0.97713633		
ex_Intercept	-1.660250500	-1.67673958		
occ_bio1_3yrs	0.958508050	0.94646554		
occ_bio2_3yrs	-0.352005625	-0.30525907		
occ_bio3_3yrs	1.552512500	1.39255795		
occ_pr_spring_3yrs	-0.132245375	-0.19300337		
occ_pr_summer_3yrs	-0.103859175	-0.17503710		
occ_pr_autumn_3yrs	1.885012750	1.61480409		
occ_pr_winter_3yrs	1.663714500	1.71835181		
occ_bio15_3yrs	-0.498071450	-0.53964252		
occ_Ibio1_3yrsE2	-0.711895425	-0.69606167		
occ_Ibio2_3yrsE2	0.464105700	0.44931287		
occ_Ibio3_3yrsE2	0.713751375	0.66907965		
occ_Ipr_spring_3yrsE2	0.207712175	0.20662797		
occ_Ipr_summer_3yrsE2	0.439200050	0.42606278		
occ_Ipr_autumn_3yrsE2	0.500264625	0.58235147		
occ_Ipr_winter_3yrsE2	-0.037167192	-0.13560596		
occ_Ibio15_3yrsE2	0.255876325	0.27519987		
colo_bio1	0.264935400	0.25021991		
colo_bio2	-0.003357925	-0.03153688		
colo_bio3	0.227445175	0.25556243		
colo_pr_spring	0.255993575	0.28092512		
colo_pr_summer	0.206317750	0.21718507		
colo_pr_autumn	0.293078225	0.26401787		
colo_pr_winter	0.090580458	0.13171926		
colo_bio15	0.046430110	0.04861843		
colo_Ibio1E2	-0.397782075	-0.39649370		
colo_Ibio2E2	0.066441370	0.08878952		
colo_Ibio3E2	0.198084500	0.20460850		
colo_Ipr_springE2	0.028728515	0.03177881		
colo_Ipr_summerE2	0.043113833	0.04626645		
colo_Ipr_autumnE2	-0.014389362	-0.04064924		
colo_Ipr_winterE2	0.028081488	0.04151751		
colo_Ibio15E2	0.023114965	0.04556341		
ex_bio1	-0.243324075	-0.24200367		

ex_bio2	0.827351400	0.82132367
ex_bio3	0.134657400	0.18199622
ex_pr_spring	0.273640450	0.26184340
ex_pr_summer	0.385368700	0.39643496
ex_pr_autumn	0.101772250	0.09445730
ex_pr_winter	-0.260352500	-0.23487573
ex_bio15	0.622561575	0.62162986
ex_Ibio1E2	0.464884675	0.47642595
ex_Ibio2E2	-0.284653950	-0.29320572
ex_Ibio3E2	-0.201324825	-0.20150348
ex_Ipr_springE2	0.146348675	0.14889201
ex_Ipr_summerE2	0.050803535	0.05710146
ex_Ipr_autumnE2	0.089210918	0.10533057
ex_Ipr_winterE2	0.398221825	0.39912408
ex_Ibio15E2	-0.102338225	-0.08770125

Quite similar.

Post-processing:

Posterior distribution for the latent occupancy state / occupancy probabilities across sites (option to do this for new data):

```
#lat_occ <- get_Z(multi_colex_all_cl_prior) # takes a while
#save(lat_occ, file = file.path("data", paste0(gsub(" ", "_", spec), "_cl_lat_occ_hs.RData"))
load(file = file.path("data", paste0(gsub(" ", "_", spec), "_cl_lat_occ_hs.RData")))
str(lat_occ) # for each site and year we have 3000 draws
```

```
'postZ' num [1:476, 1:25, 1:3000] 1.00 1.00 1.00 1.04e-07 7.09e-01 ...
```

```
lat_occ[241,3,][1:10]
```

```
[1] 0.393800972 0.190129424 0.061429473 0.022525294 0.167447760 0.039733681
[7] 0.212382152 0.027052706 0.001267152 0.094998854
```

Posterior predictions at the observed points (e.g. for use in posterior predictive checking) or for new data:

```
#post_preds <- predict_flocker(multi_colex_all_cl_prior)
```

Model comparison:

The “leave-one-out” holdouts consist of entire closure-units (single-season models), series (multi-season models).

```
#loo_mod <- loo_flocker(multi_colex_all_cl_prior, thin = 2)
# Warning: Some Pareto k diagnostic values are too high. See help('pareto-k-diagnostic') f
#save(loo_mod, file = file.path("data", paste0(gsub(" ", "_", spec), "_cl_loo_hs.RData")))
load(file = file.path("data", paste0(gsub(" ", "_", spec), "_cl_loo_hs.RData")))
loo_mod
```

Computed from 1500 by 476 log-likelihood matrix.

	Estimate	SE
elpd_loo	-23986.1	698.2
p_loo	109.4	6.2
looic	47972.2	1396.4

MCSE of elpd_loo is NA.

MCSE and ESS estimates assume MCMC draws (r_eff in [0.6, 1.1]).

Pareto k diagnostic values:

		Count	Pct.	Min. ESS
(-Inf, 0.69]	(good)	474	99.6%	117
(0.69, 1]	(bad)	2	0.4%	<NA>
(1, Inf)	(very bad)	0	0.0%	<NA>

See help('pareto-k-diagnostic') for details.

```
#loo::pareto_k_table(loo_mod)
loo::pareto_k_ids(loo_mod) # 419, 422 = sites above threshold, highly influential?
```

[1] 419 422

```
# values <- loo::pareto_k_values(loo_mod)
# infl_values <- loo::pareto_k_influence_values(loo_mod) # influence of observations on mo
# identical(values, infl_values) # TRUE
```

```
#loo_mod2 <- loo_flocker(multi_colex_all_cl_prior, thin = NULL)
#save(loo_mod2, file = file.path("data", paste0(gsub(" ", "_", spec), "_cl_loo2_hs.RData"))
```

```
load(file = file.path("data", paste0(gsub(" ", "_", spec), "_cl_loo2_hs.RData")))
loo_mod2
```

Computed from 3000 by 476 log-likelihood matrix.

	Estimate	SE
elpd_loo	-23985.5	698.3
p_loo	108.5	6.2
looic	47970.9	1396.6

MCSE of elpd_loo is NA.

MCSE and ESS estimates assume MCMC draws (r_eff in [0.5, 1.5]).

Pareto k diagnostic values:

		Count	Pct.	Min. ESS
(-Inf, 0.7]	(good)	475	99.8%	203
(0.7, 1]	(bad)	1	0.2%	<NA>
(1, Inf)	(very bad)	0	0.0%	<NA>

See help('pareto-k-diagnostic') for details.

What does this mean?

<https://mc-stan.org/loo/reference/loo-glossary.html>

- elpd = theoretically expected log pointwise predictive density for a new dataset, can be estimated using cross-validation
- elpd_loo = The Bayesian LOO estimate of out-of-sample predictive fit
 - predictive density given the data without the iths data point/series, log transformed and summed over all data points/series
- p_loo = difference between elpd_loo and non-cross-validated log posterior predictive density, how much more difficult is it to predict future data compared to observed data
 - can be interpreted as effective number of parameters, if everything is fine: $p_loo < N$ and $p_loo < p$ (=total number of parameters)
 - * warning: Some Pareto k diagnostic values are too high. See help('pareto-k-diagnostic') for details:
- looic = $-2 * elpd_loo$

- Pareto k diagnostic estimates how far an individual leave-one-out distribution is from the full distribution
 - diagnostic threshold for Pareto k depends on sample size S, loo package uses the nominal sample size S when computing the sample size specific threshold
 - Pareto k is also useful as a measure of influence of an observation. Highly influential observations have high k values.

```
#loo_comp <- loo_compare_flocker(model_list = list(multi_colex_cl, multi_colex_all_cl_prior)
#save(loo_comp, file = file.path("data", paste0(gsub(" ", "_", spec), "_cl_loo_comp_def_hs.RData"))
load(file = file.path("data", paste0(gsub(" ", "_", spec), "_cl_loo_comp_def_hs.RData")))
loo_comp
```

	elpd_diff	se_diff
model2	0.0	0.0
model1	-2.1	4.9

This tells me that model2 = model with regularizing priors is better (is this a large difference? what about the warning?)

Divergent transitions:

- If you get divergence warnings, geometry of posterior density does not allow finding a single suitable step size. Posterior density cannot be sampled well. Clear signal that something went wrong. Increasing target average acceptance probability (delta) may help.

Notes:

- there is nothing like autojags that increases iterations until chains have mixed
- you can't explicitly state which parameters to monitor. It looks like only estimates for coefficients are monitored (not the intermediate responses)